

ZARI.ai LLM Advisory Engine Technical Report

Model Architecture, Parameters, Inputs, Prompt Engineering & Safety Enforcement

1. LLM Model Specifications & Hardware Infrastructure

The ZARI.ai diagnostic advisory engine leverages Meta Llama 3.1 8B Instant (llama-3.1-8b-instant) deployed on Groq Language Processing Units (LPUs). It synthesizes retrieved ChromaDB vector store evidence into farmer-friendly, evidence-grounded treatment plans for Pakistani agriculture.

Primary Large Language Model Parameters

Model Name:	Meta Llama 3.1 8B Instant (llama-3.1-8b-instant)
Developer / Provider:	Meta AI / Groq API Infrastructure
Total Parameter Count:	8.0 Billion Parameters (8 x 10 ⁹ params)
Architecture Type:	Auto-regressive Transformer Decoder with Grouped Query Attention (GQA)
Context Window Size:	128,000 Tokens (128k context length)
Inference Acceleration:	Groq LPU (Language Processing Unit) Tensor Hardware
Generation Speed:	~750 Tokens / Second (Sub-second latency: 350-500 ms)
Fallback Generator:	Deterministic Evidence Synthesis Engine (100% offline fallback)

2. LLM Inputs & Data Payloads

The LLM generator receives 5 structured inputs from the Vision AI pipeline and RAG retriever:

Input Field	Data Type	Sample Value	Function & Description
disease_class	string	Tomato_Late_Blight	Canonical disease class identified by Vision AI
confidence	float	0.9845 (98.5%)	Vision classifier softmax confidence score
uncertainty	float	0.1245	Evidential Deep Learning (EDL) uncertainty $u = K / S$
language	string	ur / ps / en	Target farmer output language (Urdu, Pashto, English)
retrieved_chunks	list[dict]	Top-4 ChromaDB Chunks	Ground-truth evidence chunks containing IPM guidelines

3. Prompt Engineering & System Directives

The prompt template enforces strict grounding and zero-hallucination guardrails:

- Zero Invented Dosages: The LLM is strictly prohibited from inventing chemical spray dosages (ml/L or g/L).
- Zero Invented PHI Days: Pre-Harvest Intervals (PHI) must refer farmers to official product label instructions.
- IPM Order Enforcement: Recommendations must strictly follow Cultural -> Biological -> Chemical controls.
- Viral Pathogen Safety: For viral infections (e.g. Tomato Leaf Curl), synthetic fungicides are forbidden; vector control is mandated.
- Quarantine Protocols: High-caution quarantine warnings for severe regional crop threats.

4. Latency Disambiguation: Online vs Offline Pipeline

Vision Inference Latency:	3.24 ms CUDA inference time
ChromaDB Retrieval Latency:	5.32 ms HNSW search time
Offline Fallback Synthesis Latency:	0.86 ms deterministic formatting time
TOTAL OFFLINE EDGE LATENCY:	9.40 ms (3.24ms Vision + 5.32ms RAG + 0.86ms Fallback)
Groq Llama 3.1 8B API Call:	380 ms mean network + generation latency
TOTAL ONLINE CLOUD LATENCY:	389 ms (3.24ms Vision + 5.32ms RAG + 380ms Groq API)